Lesson 10 material: Understanding Psychological Measurement

4.1: Prelude to Psychological Measurement

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Researchers Tara MacDonald and Alanna Martineau were interested in the effect of female university students' moods on their intentions to have unprotected sexual intercourse (MacDonald & Martineau, 2002)[1]. In a carefully designed empirical study, they found that being in a negative mood increased intentions to have unprotected sex—but only for students who were low in self-esteem. Although there are many challenges involved in conducting a study like this, one of the primary ones is the measurement of the relevant variables. In this study, the researchers needed to know whether each of their participants had high or low self-esteem, which of course required measuring their self-esteem. They also needed to be sure that their attempt to put people into a negative mood (by having them think negative thoughts) was successful, which required measuring their moods. Finally, they needed to see whether self-esteem and mood were related to participants' intentions to have unprotected sexual intercourse, which required measuring these intentions.

Do You Feel You Are a Person of Worth?

The Rosenberg Self-Esteem Scale (Rosenberg, 1989)[2] is one of the most common measures of self-esteem and the one that MacDonald and Martineau used in their study. Participants respond to each of the 10 items that follow with a rating on a 4-point scale: *Strongly Agree*, *Agree*, *Disagree*, *Strongly Disagree*. Score Items 1, 2, 4, 6, and 7 by assigning 3 points for each *Strongly Agree* response, 2 for each *Agree*, 1 for each *Disagree*, and 0 for each *Strongly Disagree*. Reverse the scoring for Items 3, 5, 8, 9, and 10 by assigning 0 points for each *Strongly Agree*, 1 point for each *Agree*, and so on. The overall score is the total number of points.

1. I feel that I'm a person of worth, at least on an equal plane with others.
2. I feel that I have a number of good qualities.
3. All in all, I am inclined to feel that I am a failure.
4. I am able to do things as well as most other people.
5. I feel I do not have much to be proud of.
6. I take a positive attitude toward myself.
7. On the whole, I am satisfied with myself.
8. I wish I could have more respect for myself.
9. I certainly feel useless at times.
10. At times I think I am no good at all.

1. MacDonald, T. K., & Martineau, A. M. (2002). Self-esteem, mood, and intentions to use condoms: When does low self-esteem lead to risky health behaviors? *Journal of Experimental Social Psychology, 38*, 299–306. ↵
2. Rosenberg, M. (1989). *Society and the adolescent self-image* (rev. ed.). Middletown, CT: Wesleyan University Press. ↵

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4.2: Understanding Psychological Measurement

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Learning Objectives

1. Define measurement and give several examples of measurement in psychology.
2. Explain what a psychological construct is and give several examples.
3. Distinguish conceptual from operational definitions, give examples of each, and create simple operational definitions.
4. Distinguish the four levels of measurement, give examples of each, and explain why this distinction is important.

What Is Measurement?

Measurement is the assignment of scores to individuals so that the scores represent some characteristic of the individuals. This very general definition is consistent with the kinds of measurement that everyone is familiar with—for example, weighing oneself by stepping onto a bathroom scale, or checking the internal temperature of a roasting turkey using a meat thermometer. It is also consistent with measurement in the other sciences. In physics, for example, one might measure the potential energy of an object in Earth's gravitational field by finding its mass and height (which of course requires measuring *those* variables) and then multiplying them together along with the gravitational acceleration of Earth (9.8 m/s2). The result of this procedure is a score that represents the object's potential energy.

This general definition of measurement is consistent with measurement in psychology too. (Psychological measurement is often referred to as **psychometrics**.) Imagine, for example, that a cognitive psychologist wants to measure a person's working memory capacity—their ability to hold in mind and think about several pieces of information all at the same time. To do this, she might use a backward digit span task, in which she reads a list of two digits to the person and asks them to repeat them in reverse order. She then repeats this several times, increasing the length of the list by one digit each time, until the person makes an error. The length of the longest list for which the person responds correctly is the score and represents their working memory capacity. Or imagine a clinical psychologist who is interested in how depressed a person is. He administers the Beck Depression Inventory, which is a 21-item self-report questionnaire in which the person rates the extent to which they have felt sad, lost energy, and experienced other symptoms of depression over the past 2 weeks. The sum of these 21 ratings is the score and represents the person's current level of depression.

The important point here is that measurement does not require any particular instruments or procedures. What it *does* require is *some* systematic procedure for assigning scores to individuals or objects so that those scores represent the characteristic of interest.

Psychological Constructs

Many variables studied by psychologists are straightforward and simple to measure. These include age, height, weight, and birth order. You can ask people how old they are and be reasonably sure that they know and will tell you. Although people might not know or want to tell you how much they weigh, you can have them step onto a bathroom scale. Other variables studied by psychologists—perhaps the majority—are not so straightforward or simple to measure. We cannot accurately assess people's level of intelligence by looking at them, and we certainly cannot put their self-esteem on a bathroom scale. These kinds of variables are called **constructs** (pronounced *CON-structs*) and include personality traits (e.g., extraversion), emotional states (e.g., fear), attitudes (e.g., toward taxes), and abilities (e.g., athleticism).

Psychological constructs cannot be observed directly. One reason is that they often represent *tendencies* to think, feel, or act in certain ways. For example, to say that a particular university student is highly extraverted does not necessarily mean that she is behaving in an extraverted way right now. In fact, she might be sitting quietly by herself, reading a book. Instead, it means that she has a general tendency to behave in extraverted ways (e.g., being outgoing, enjoying social interactions) across a variety of situations. Another reason psychological constructs cannot be observed directly is that they often involve internal processes. Fear, for example, involves the activation of certain central and peripheral nervous system structures, along with certain kinds of thoughts, feelings, and behaviors—none of which is necessarily obvious to an outside observer. Notice also that neither extraversion nor fear "reduces to" any particular thought, feeling, act, or physiological structure or process. Instead, each is a kind of summary of a complex set of behaviors and internal processes.

The Big Five

The Big Five is a set of five broad dimensions that capture much of the variation in human personality. Each of the Big Five can even be defined in terms of six more specific constructs called "facets" (Costa & McCrae, 1992)[1].

Figure: The Big Five Personality Dimensions

The **conceptual definition** of a psychological construct describes the behaviors and internal processes that make up that construct, along with how it relates to other variables. For example, a conceptual definition of neuroticism (another one of the Big Five) would be that it is people's tendency to experience negative emotions such as anxiety, anger, and sadness across a variety of situations. This definition might also include that it has a strong genetic component, remains fairly stable over time, and is positively correlated with the tendency to experience pain and other physical symptoms.

Students sometimes wonder why, when researchers want to understand a construct like self-esteem or neuroticism, they do not simply look it up in the dictionary. One reason is that many scientific constructs do not have counterparts in everyday language (e.g., working memory capacity). More important, researchers are in the business of developing definitions that are more detailed and precise—and that more accurately describe the way the world is—than the informal definitions in the dictionary. As we will see, they do this by proposing conceptual definitions, testing them empirically, and revising them as necessary. Sometimes they throw them out altogether. This is why the research literature often includes different conceptual definitions of the same

construct. In some cases, an older conceptual definition has been replaced by a newer one that fits and works better. In others, researchers are still in the process of deciding which of various conceptual definitions is the best.

Operational Definitions

An **operational definition** is a definition of a variable in terms of precisely how it is to be measured. These measures generally fall into one of three broad categories. **Self-report measures** are those in which participants report on their own thoughts, feelings, and actions, as with the Rosenberg Self-Esteem Scale (Rosenberg, 1965)[2]. **Behavioral measures** are those in which some other aspect of participants' behavior is observed and recorded. This is an extremely broad category that includes the observation of people's behavior both in highly structured laboratory tasks and in more natural settings. A good example of the former would be measuring working memory capacity using the backward digit span task. A good example of the latter is a famous operational definition of physical aggression from researcher Albert Bandura and his colleagues (Bandura, Ross, & Ross, 1961)[3]. They let each of several children play for 20 minutes in a room that contained a clown-shaped punching bag called a Bobo doll. They filmed each child and counted the number of acts of physical aggression the child committed. These included hitting the doll with a mallet, punching it, and kicking it. Their operational definition, then, was the number of these specifically defined acts that the child committed during the 20-minute period. Finally, **physiological measures** are those that involve recording any of a wide variety of physiological processes, including heart rate and blood pressure, galvanic skin response, hormone levels, and electrical activity and blood flow in the brain.

For any given variable or construct, there will be multiple operational definitions. Stress is a good example. A rough conceptual definition is that stress is an adaptive response to a perceived danger or threat that involves physiological, cognitive, affective, and behavioral components. But researchers have operationally defined it in several ways. The Social Readjustment Rating Scale (Holmes & Rahe, 1967)[4] is a self-report questionnaire on which people identify stressful events that they have experienced in the past year and assigns points for each one depending on its severity. For example, a man who has been divorced (73 points), changed jobs (36 points), and had a change in sleeping habits (16 points) in the past year would have a total score of 125. The Hassles and Uplifts Scale (Delongis, Coyne, Dakof, Folkman & Lazarus, 1982) [5] is similar but focuses on everyday stressors like misplacing things and being concerned about one's weight. The Perceived Stress Scale (Cohen, Kamarck, & Mermelstein, 1983) [6] is another self-report measure that focuses on people's feelings of stress (e.g., "How often have you felt nervous and stressed?"). Researchers have also operationally defined stress in terms of several physiological variables including blood pressure and levels of the stress hormone cortisol.

When psychologists use multiple operational definitions of the same construct—either within a study or across studies—they are using **converging operations**. The idea is that the various operational definitions are "converging" or coming together on the same construct. When scores based on several different operational definitions are closely related to each other and produce similar patterns of results, this constitutes good evidence that the construct is being measured effectively and that it is useful. The various measures of stress, for example, are all correlated with each other and have all been shown to be correlated with other variables such as immune system functioning (also measured in a variety of ways) (Segerstrom & Miller, 2004)[7]. This is what allows researchers eventually to draw useful general conclusions, such as "stress is negatively correlated with immune system functioning," as opposed to more specific and less useful ones, such as "people's scores on the Perceived Stress Scale are negatively correlated with their white blood counts."

Levels of Measurement

The psychologist S. S. Stevens suggested that scores can be assigned to individuals in a way that communicates more or less quantitative information about the variable of interest (Stevens, 1946)[8]. For example, the officials at a 100-m race could simply rank order the runners as they crossed the finish line (first, second, etc.), or they could time each runner to the nearest tenth of a second using a stopwatch (11.5 s, 12.1 s, etc.). In either case, they would be measuring the runners' times by systematically assigning scores to represent those times. But while the rank ordering procedure communicates the fact that the second-place runner took longer to finish than the first-place finisher, the stopwatch procedure also communicates *how much* longer the second-place finisher took. Stevens actually suggested four different **levels of measurement** (which he called "scales of measurement") that correspond to four types of information that can be communicated by a set of scores, and the statistical procedures that can be used with the information.

The **nominal level** of measurement is used for categorical variables and involves assigning scores that are category labels. Category labels communicate whether any two individuals are the same or different in terms of the variable being measured. For example, if you ask your participants about their marital status, you are engaged in nominal-level measurement. Or if you ask your participants to indicate which of several ethnicities they identify themselves with, you are again engaged in nominal-level measurement. The essential point about nominal scales is that they do not imply any ordering among the responses. For example, when classifying people according to their favorite color, there is no sense in which green is placed "ahead of" blue. Responses are merely categorized. Nominal scales thus embody the lowest level of measurement[9].

The remaining three levels of measurement are used for quantitative variables. The **ordinal level** of measurement involves assigning scores so that they represent the rank order of the individuals. Ranks communicate not only whether any two individuals are the same or different in terms of the variable being measured but also whether one individual is higher or lower on that variable. For example, a researcher wishing to measure consumers' satisfaction with their microwave ovens might ask them to specify their feelings as either "very dissatisfied," "somewhat dissatisfied," "somewhat satisfied," or "very satisfied." The items in this scale are ordered, ranging from least to most satisfied. This is what distinguishes ordinal from nominal scales. Unlike nominal scales, ordinal scales allow comparisons of the degree to which two individuals rate the variable. For example, our satisfaction ordering makes it meaningful to assert that one person is more satisfied than another with their microwave ovens. Such an assertion reflects the first person's use of a verbal label that comes later in the list than the label chosen by the second person.

On the other hand, ordinal scales fail to capture important information that will be present in the other levels of measurement we examine. In particular, the difference between two levels of an ordinal scale cannot be assumed to be the same as the difference between two other levels (just like you cannot assume that the gap between the runners in first and second place is equal to the gap between the runners in second and third place). In our satisfaction scale, for example, the difference between the responses "very dissatisfied" and "somewhat dissatisfied" is probably not equivalent to the difference between "somewhat dissatisfied" and "somewhat satisfied." Nothing in our measurement procedure allows us to determine whether the two differences reflect the same difference in psychological satisfaction. Statisticians express this point by saying that the differences between adjacent scale values do not necessarily represent equal intervals on the underlying scale giving rise to the measurements. (In our case, the underlying scale is the true feeling of satisfaction, which we are trying to measure.)

The **interval level** of measurement involves assigning scores using numerical scales in which intervals have the same interpretation throughout. As an example, consider either the Fahrenheit or Celsius temperature scales. The difference between 30 degrees and 40 degrees represents the same temperature difference as the difference between 80 degrees and 90 degrees. This is because each 10-degree interval has the same physical meaning (in terms of the kinetic energy of molecules).

Interval scales are not perfect, however. In particular, they do not have a true zero point even if one of the scaled values happens to carry the name "zero." The Fahrenheit scale illustrates the issue. Zero degrees Fahrenheit does not represent the complete absence of temperature (the absence of any molecular kinetic energy). In reality, the label "zero" is applied to its temperature for quite accidental reasons connected to the history of temperature measurement. Since an interval scale has no true zero point, it does not make sense to compute ratios of temperatures. For example, there is no sense in which the ratio of 40 to 20 degrees Fahrenheit is the same as the ratio of 100 to 50 degrees; no interesting physical property is preserved across the two ratios. After all, if the "zero" label were applied at the temperature that Fahrenheit happens to label as 10 degrees, the two ratios would instead be 30 to 10 and 90 to 40, no longer the same! For this reason, it does not make sense to say that 80 degrees is "twice as hot" as 40 degrees. Such a claim would depend on an arbitrary decision about where to "start" the temperature scale, namely, what temperature to call zero (whereas the claim is intended to make a more fundamental assertion about the underlying physical reality).

In psychology, the intelligence quotient (IQ) is often considered to be measured at the interval level. While it is technically possible to receive a score of 0 on an IQ test, such a score would not indicate the complete absence of IQ. Moreover, a person with an IQ score of 140 does not have twice the IQ of a person with a score of 70. However, the difference between IQ scores of 80 and 100 is the same as the difference between IQ scores of 120 and 140.

Finally, the **ratio level** of measurement involves assigning scores in such a way that there is a true zero point that represents the complete absence of the quantity. Height measured in meters and weight measured in kilograms are good examples. So are counts of discrete objects or events such as the number of siblings one has or the number of questions a student answers correctly on an exam. You can think of a ratio scale as the three earlier scales rolled up in one. Like a nominal scale, it provides a name or category for each object (the numbers serve as labels). Like an ordinal scale, the objects are ordered (in terms of the ordering of the numbers). Like an interval scale, the same difference at two places on the scale has the same meaning. However, in addition, the same ratio at two places on the scale also carries the same meaning (see Table).

The Fahrenheit scale for temperature has an arbitrary zero point and is therefore not a ratio scale. However, zero on the Kelvin scale is absolute zero. This makes the Kelvin scale a ratio scale. For example, if one temperature is twice as high as another as measured on the Kelvin scale, then it has twice the kinetic energy of the other temperature.

Another example of a ratio scale is the amount of money you have in your pocket right now (25 cents, 50 cents, etc.). Money is measured on a ratio scale because, in addition to having the properties of an interval scale, it has a true zero point: if you have zero money, this actually implies the absence of money. Since money has a true zero point, it makes sense to say that someone with 50 cents has twice as much money as someone with 25 cents.

Stevens's levels of measurement are important for at least two reasons. First, they emphasize the generality of the concept of measurement. Although people do not normally think of categorizing or ranking individuals as measurement, in fact, they are as long as they are done so that they represent some characteristic of the individuals. Second, the levels of measurement can serve as a rough guide to the statistical procedures that can be used with the data and the conclusions that can be drawn from them. With nominal-level measurement, for example, the only available measure of central tendency is the mode. With ordinal-level measurement, the median or mode can be used as indicators of central tendency. Interval and ratio-level measurement are typically considered the most desirable because they permit for any indicators of central tendency to be computed (i.e., mean, median, or mode). Also, ratio-level measurement is the only level that allows meaningful statements about ratios of scores. Once again, one cannot say that someone with an IQ of 140 is twice as intelligent as someone with an IQ of 70 because IQ is measured at the interval level, but one can say that someone with six siblings has twice as many as someone with three because number of siblings is measured at the ratio level.

Table Summary of Levels of Measurements

Level of Measurement	Category labels	Rank order	Equal intervals	True zero
NOMINAL	X			
ORDINAL	X	X		
INTERVAL	X	X	X	
RATIO	X	X	X	X

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